

TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN
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APPLICATIONS OF GRAPH DATA STRUCTURE

Subject: Data structures and algorithms

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APPLICATIONS OF GRAPH DATA STRUCTURE

1. Definition

- A graph data structure is a collection of nodes that have data and are connected to other nodes.

2. Operations for working with graph data type structures

- Commonly used operations with the most common graph types include the following:
 - Checks whether the element is present in the graph.
 - Browse the graph.
 - Add elements (vertices, edges) to the graph.
 - Find the path from one vertex to another.

3. Applications of graphs

- Here we apply graph structure to Facebook.
- The Facebook graph is the collection of entities and their relationships on Facebook. The entities are the nodes and the relationships are the edges. ... Edges between nodes represent friendships, check-ins, tags, relationships, ownership, attributes, etc. Both nodes and edges have metadata associated with them.
- Facebook's friend recommendation algorithm uses graph theory. Facebook is an example of an undirected graph.
- Facebook's Graph API is perhaps the best example of application of graphs to real life problems. The Graph API is a revolution in large-scale data provision.
- On The Graph API, everything is a vertice or node. This are entities such as Users, Pages, Places, Groups, Comments, Photos, Photo Albums, Stories, Videos, Notes, Events and so forth. Anything that has properties that store data is a vertice.
- And every connection or relationship is an edge. This will be something like a User posting a Photo, Video or Comment etc., a User updating their profile with a their Place of birth, a relationship status Users, a User liking a Friend's Photo etc.
- The Graph API uses this collections of vertices and edges (essentially graph data structures) to store its data. The Graph API is also a GraphQL API. This is the language it uses to build and query the schema.
- The Graph API has come into some problems because of it's ability to obtain unusually rich info about user's friends.